# Anas Barakat

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## RESEARCH PROFILE

Foundations and algorithms for optimization and learning in sequential and strategic decision making, with a forward-looking agenda on reliable AI post-training and learning and incentive design for interacting AI agents.

**Research themes:**

- **Policy optimization with richer objectives:** general utility reinforcement learning, policy gradient methods, nonlinear objectives, and LLM post-training; toward reliable post-training and sequential calibration.
- **Multi-agent learning in strategic and dynamic environments:** learning dynamics and equilibrium learning in static, Markov, and dynamical games, learning from preference and limited feedback; toward learning and incentive design in agentic markets.

**Foundations:** optimization, reinforcement learning, game theory, online learning, stochastic approximation.

## ACADEMIC APPOINTMENTS

**Singapore University of Technology and Design** Oct. 2024–present
Research Fellow; hosts: Prof. Georgios Piliouras and Prof. Antonios Varvitsiotis

**ETH Zurich, Department of Computer Science** Feb. 2022–Aug. 2024
Foundations of Data Science Postdoctoral Fellow; host: Prof. Niao He

## EDUCATION

**Institut Polytechnique de Paris, Télécom Paris**, Paris, France 2018–2021
Ph.D. in Applied Mathematics and Computer Science
Advisors: Prof. Pascal Bianchi and Prof. Walid Hachem

**Université Paris-Saclay**, Paris, France 2017–2018
M.Sc. in Data Science, highest honors

**Télécom Paris**, Paris, France 2015–2018
M.Sc. in Applied Mathematics and Computer Science

## SELECTED RECENT PUBLICATIONS

1. **Anas Barakat**. **Preprint 2026**
*Learning in Continuous Games from Pairwise Preference Feedback.*

2. **Anas Barakat**, Souradip Chakraborty, Khushbu Pahwa, Amrit Singh Bedi. **NeurIPS 2026**
*Prompt Interference in LLM Post-Training: Why Pass@k Optimization Can Degrade Pass@1.*

3. **Anas Barakat**, Andreas Kontogiannis, Vasilis Pollatos, Ioannis Panageas, Antonios Varvitsiotis. **NeurIPS 2026**
*Online Learning on Hidden-Convex Losses via Algorithmic Equivalence: Optimal Regret, Geometric Barrier, and Bandit Feedback.*

4. **Anas Barakat**, John Lazarsfeld, Georgios Piliouras, Antonios Varvitsiotis. **NeurIPS 2026**
*Online Learning in Stabilized Linear Dynamical Games with Adversarial Disturbances.*

5. John Lazarsfeld, **Anas Barakat**, Georgios Piliouras, Antonios Varvitsiotis, Andre Wibisono. **NeurIPS 2026**
*When and Why is Optimistic Multiplicative Weights Slow? The Geometry of Energy Dissipation.*

6. **Anas Barakat**, Ioannis Panageas, Antonios Varvitsiotis. **AISTATS 2026**
*Convex Markov Games and Beyond: New Proof of Existence, Characterization and Learning Algorithms for Nash Equilibria.*

7. **Anas Barakat**, Souradip Chakraborty, Peihong Yu, Pratap Tokekar, Amrit Singh Bedi. **NeurIPS 2025**
*On the Global Optimality of Policy Gradient Methods in General Utility Reinforcement Learning.*

## PUBLICATIONS & PREPRINTS

Names of student co-authors I have previously supervised and mentored are underlined.

### Preprints

(1) **Anas Barakat**. **Preprint 2026**
*Learning in Continuous Games from Pairwise Preference Feedback.*

(2) John Lazarsfeld, **Anas Barakat**, Jacob Abernethy, Vidya Muthukumar, Georgios Piliouras, Andre Wibisono, Juba Ziani. **Preprint 2026**
*Linear Last-Iterate Convergence Rates for Optimistic Learning in Harmonic Games.*

### Refereed Conference Publications

(C1) **Anas Barakat**, Souradip Chakraborty, Khushbu Pahwa, Amrit Singh Bedi. **NeurIPS 2026**
*Prompt Interference in LLM Post-Training: Why Pass@k Optimization Can Degrade Pass@1.* Advances in Neural Information Processing Systems, 2026.

(C2) **Anas Barakat**, Andreas Kontogiannis, Vasilis Pollatos, Ioannis Panageas, Antonios Varvitsiotis. **NeurIPS 2026**
*Online Learning on Hidden-Convex Losses via Algorithmic Equivalence: Optimal Regret, Geometric Barrier, and Bandit Feedback.* Advances in Neural Information Processing Systems, 2026.

(C3) John Lazarsfeld, **Anas Barakat**, Georgios Piliouras, Antonios Varvitsiotis, Andre Wibisono. **NeurIPS 2026**
*When and Why is Optimistic Multiplicative Weights Slow? The Geometry of Energy Dissipation.* Advances in Neural Information Processing Systems, 2026.

(C4) **Anas Barakat**, John Lazarsfeld, Georgios Piliouras, Antonios Varvitsiotis. **NeurIPS 2026**
*Online Learning in Stabilized Linear Dynamical Games with Adversarial Disturbances.* Advances in Neural Information Processing Systems, 2026.

(C5) **Anas Barakat**, Ioannis Panageas, Antonios Varvitsiotis. **AISTATS 2026**
*Convex Markov Games and Beyond: New Proof of Existence, Characterization and Learning Algorithms for Nash Equilibria.* International Conference on Artificial Intelligence and Statistics, 2026.

(C6) **Anas Barakat**, Souradip Chakraborty, Peihong Yu, Pratap Tokekar, Amrit Singh Bedi. **NeurIPS 2025**
*On the Global Optimality of Policy Gradient Methods in General Utility Reinforcement Learning.* Advances in Neural Information Processing Systems, 2025.

(C7) Kimon Protopapas, **Anas Barakat**. **NeurIPS 2024**
*Policy Mirror Descent with Lookahead.* Advances in Neural Information Processing Systems, 2024.

(C8) Pragnya Alatur, **Anas Barakat**[*], Niao He. **CDC 2024**
*Independent Policy Mirror Descent for Markov Potential Games: Scaling to Large Number of Players.* IEEE Conference on Decision and Control, 2024. [*]Corresponding author.

(C9) Philip Jordan, **Anas Barakat**, Niao He. **AISTATS 2024**
*Independent Learning in Constrained Markov Potential Games.*
International Conference on Artificial Intelligence and Statistics, 2024.

(C10) Jiduan Wu, **Anas Barakat**, Ilyas Fatkhullin, Niao He. **CDC 2023**
*Learning Zero-Sum Linear Quadratic Games with Improved Sample Complexity.*
IEEE Conference on Decision and Control, 2023.

(C11) **Anas Barakat**, Ilyas Fatkhullin, Niao He. **ICML 2023**
*Reinforcement Learning with General Utilities: Simpler Variance Reduction and Large State-Action Space.* International Conference on Machine Learning, 2023.

(C12) Ilyas Fatkhullin, **Anas Barakat**, Anastasia Kireeva, Niao He. **ICML 2023**
*Stochastic Policy Gradient Methods: Improved Sample Complexity for Fisher-non-degenerate Policies.* International Conference on Machine Learning, 2023.

(C13) **Anas Barakat**, Pascal Bianchi, Julien Lehmann. **AISTATS 2022**
*Analysis of a Target-Based Actor-Critic Algorithm with Linear Function Approximation.*
International Conference on Artificial Intelligence and Statistics, 2022.

(C14) **Anas Barakat**, Pascal Bianchi. **ACML 2020**
*Convergence Rates of a Momentum Algorithm with Bounded Adaptive Step Size for Nonconvex Optimization.* Asian Conference on Machine Learning, 2020.

## Refereed Journal Publications

(J1) Olivier Lepel, **Anas Barakat**. **TMLR 2026**
*Policy Gradients for Cumulative Prospect Theory in Reinforcement Learning.*
Transactions on Machine Learning Research, 2026.

(J2) **Anas Barakat**, Wayne Lin, John Lazarsfeld, Antonios Varvitsiotis. **TMLR 2026**
*Optimistic Online Learning in Symmetric Cone Games.*
Transactions on Machine Learning Research, 2026.

(J3) Jiduan Wu, **Anas Barakat**, Ilyas Fatkhullin, Niao He. **SICON 2025**
*Learning Zero-Sum Linear Quadratic Games with Improved Sample Complexity and Last-Iterate Convergence.* SIAM Journal on Control and Optimization, 2025.

(J4) **Anas Barakat**, Pascal Bianchi, Walid Hachem, Sholom Schechtman. **EJS 2021**
*Stochastic Optimization with Momentum: Convergence, Fluctuations, and Traps Avoidance.*
Electronic Journal of Statistics, 15(2):3892–3947, 2021.

(J5) **Anas Barakat**, Pascal Bianchi. **SIOPT 2021**
*Convergence and Dynamical Behavior of the Adam Algorithm for Non-Convex Stochastic Optimization.* SIAM Journal on Optimization, 31(1):244–274, 2021.

## AWARDS AND SCHOLARSHIPS

- ETH Zurich Foundations of Data Science Postdoctoral Fellowship 2022–2024
- DAAD Postdoctoral Networking Tour in AI (Postdoc-NeT-AI) Fellow Nov. 2023
- Dodu Prize, French Society of Applied and Industrial Mathematics (SMAI) Sep. 2021
  Best communication by a young researcher at the Optimization and Decision annual meeting
- Mines-Télécom Institute (IMT) Ph.D. Scholarship 2018–2021
- Moroccan Government Merit Scholarship 2015–2018
- Agency for French Education Abroad (AEFE) "Excellence-Major" Scholarship 2013–2015

## STUDENT RESEARCH MENTORING

Led supervision of six master's/semester research projects; five led to refereed publications.

- Olivier Lepel — Master's thesis, Mar.–Sep. 2024. **TMLR 2026**
  *Beyond Expected Returns: A Policy Gradient Algorithm for Cumulative Prospect Theoretic Reinforcement Learning.*
- Kimon Protopapas — Master's semester project, Sep. 2023–Jan. 2024. **NeurIPS 2024**
  *Policy Mirror Descent with Lookahead.*
- Philip Jordan — Master's thesis, May–Dec. 2023. **AISTATS 2024**
  *Independent Learning in Constrained Markov Potential Games.*
- Jiduan Wu — Master's thesis, Oct. 2022–Mar. 2023. **CDC 2023 · SICON 2025**
  *Learning Zero-Sum Linear Quadratic Games with Improved Sample Complexity and Last-Iterate Convergence.*
- Harish Rajagopal — Master's thesis, Mar.–Sep. 2022.
  *Multistage Step Size Scheduling for Minimax Problems.*
- Julien Lehmann — Master's thesis, May–Oct. 2021. **AISTATS 2022**
  *Analysis of a Target-Based Actor-Critic Algorithm with Linear Function Approximation.*

## SELECTED INVITED TALKS AND SEMINARS

- Invited talk, 5th Symposium on Machine Learning and Dynamical Systems 10 Feb. 2026
  Kyoto University, Kyoto, Japan
- Invited talk, ICCOPT 2025, University of Southern California, Los Angeles, USA 24 Jul. 2025
- Invited talk, Learning Theory and Applications Workshop 29 Apr. 2025
  College of Computing and Data Science, Nanyang Technological University, Singapore
  Host: Themis Gouleakis.

- Invited talk (online), Finance and RL Talks, FinRL core team & Melwy open AI lab — 21 Apr. 2025
  Host: Mostapha Benhenda.
- Invited talk, Fourth Symposium on Machine Learning and Dynamical Systems — 11 Jul. 2024
  Fields Institute, Toronto, Canada.
- Invited talk, ICCOPT 2022, Lehigh University, Bethlehem, USA — 28 Jul. 2022
- Invited talk (online), 14th CMStatistics International Conference — 20 Dec. 2021
  Session: "Dynamical systems in machine learning" organized by Anna Korba
  King's College London, United Kingdom
- Invited seminar, *Image, Optimization and Probabilities* research group — 15 Oct. 2020
  Bordeaux Institute of Mathematics (IMB), Bordeaux, France
- Invited talk (online), 2nd Symposium on Machine Learning and Dynamical Systems — 21 Sep. 2020
  Fields Institute for Research in Mathematical Sciences, online

## TEACHING

- **Singapore University of Technology and Design**, Singapore
  - Co-designed and co-taught a new graduate course:
    *Topics in Games, Learning and Optimization* — Fall 2025
- **ETH Zurich**, Zurich, Switzerland
  - Head Teaching Assistant for 'Foundations of Reinforcement Learning' — Spring 2023
  - Instructor for two lectures in 'Optimization for Data Science' — Spring 2023
  - Instructor for a lecture on 'Value-Based Methods' in 'Foundations of RL' — Spring 2024
  - Coordinator for the seminar course 'Advanced Topics in Machine Learning' — Fall 2022, 2023
  - Teaching Assistant for 'Optimization for Data Science' and 'Foundations of RL' — Spring 2024
- **Télécom Paris**, Paris, France
  - Teaching Assistant for 'Optimization for Machine Learning' — Fall 2018, 2019, 2020
  - Instructor for discrete Markov chains in 'Probabilities' — Fall 2018, 2019
  - Teaching Assistant for 'Probabilities', 'Statistics', 'Machine Learning' — Fall 2018, 2019
- **École Polytechnique**, Paris, France
  - Tutor: Design and Analysis of Algorithms; Advanced Programming ($\sim$ 20 hours) — Spring 2017

## ACADEMIC SERVICE

- **Area Chair:** ICLR 2027, ICML 2027.
- **Reviewer distinctions:** NeurIPS 2026 Top Reviewer; ICML 2026 Gold Reviewer (Top 25%); AISTATS 2022 Top 10% Reviewer.
- **Conference reviewing:** NeurIPS (2022, 2023, 2025, 2026), ICML (2024–2026), ICLR (2025–2026), AISTATS (2022, 2024, 2026), EC 2026.
- **Journal reviewing:** JMLR, TMLR, Mathematical Programming, SIOPT, JOTA, IEEE TAC, Stochastic Systems, Systems & Control Letters, MCSS, IEEE L-CSS.
- **Organization of workshops, summer schools and conferences:**
  - Co-organizer, Multi-Agent RL EPFL–ETHZ Summer School 2024; co-developed successful CHF 20,000 funding proposal.
  - ICCOPT 2022, session organizer and chair: *Policy Gradient and Actor-Critic Methods: Theoretical Analysis and New Opportunities*, in the Optimization for Data Science and ML cluster.
  - Workshop organizer: RL workshop for industrial partners (Airbus Defence & Space, Engie, Idemia, Safran and Valeo) of the Data Science & AI for Digitalized Industry & Services research and teaching chair.